

8 (short)

- (a) What is a static hazard in a combinational digital circuit? [3]
- (b) Identify potential static 0 or static 1 hazards when the four-variable function Z is implemented where $Z = \sum(1, 3, 5, 7, 8, 9, 12, 13)$. Take A, B, C and D to be the four variables with A being the most significant. How can these hazards be avoided? [7]

9 (long) A 4 bit synchronous binary up counter is to be designed using JK flip-flops.

- (a) Write down the complete truth table. [6]
- (b) Construct the Karnaugh maps and derive the logic expressions for the J and K inputs. [6]
- (c) Draw the full circuit diagram. Explain how carry in and carry out signals could be implemented in this design. [12]
- (d) Draw the outputs Q_n (where $n = 1, 2, 3, 4$) corresponding to 10 clock cycles. Assume a 10 MHz clock frequency. [6]